GEOMETRY · SIMILARITY

Dilation & Similarity Lab

Explore & Reflect

Name _____ Date _____ Period _____

Open **Dilation & Similarity Lab** on Mr. K's Classroom: Geometry → Similarity → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. If every side of a triangle doubles, what happens to its area?

_____

_____

2 Explore use the tool and record what you see

Dilate △ABC with A(1, 1), B(4, 1), C(1, 3)

Original: AB = 3, perimeter ≈ 8.61, area = 3. Change k and record.

k	A′B′	Perimeter ratio	Area ratio	Image area
2				
3				
$\frac{1}{2}$				
$\frac{3}{2}$				

What stays the same?

	Preimage	Image (k = 2)
Angles		
Side ratios AB : AC		

3 Write the rule fill in each blank from what you observed

1. A dilation multiplies every length by _____.
2. Angles in the image are _____ the angles in the preimage.
3. Perimeter is multiplied by _____ and area by _____.
4. $k > 1$ makes an _____. $0 < k < 1$ makes a _____.
5. The center of dilation is the one point that _____.

4 Check no tool: show your work

1. A segment of length 2.5 is dilated by k = 4. How long is the image?

2. Similar triangles have corresponding sides 6 and 9. Find the scale factor from small to large.

3. A figure with area 20 is dilated by k = 3. Find the new area.

4. Dilate (2, −4) by $\frac{1}{2}$ about the origin.

5 Reflect

Why does the area grow by k^2 instead of k?